Final Assignment Tutorial

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Lecturer



Name: **Shunsuke Kamiya**

Who am I:

- a post-doc researcher at Matsuo Lab
- Lead developer of GCI World educational materials (with Aki)

Interest:

- Dynamical systems x Machine learning
- AIs that think deeply

Hobby:

- Listening to music (of any kind: classical, jazz, 90’s R&B, hip-hop, J-pop, etc)

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Contents

- Overview of Final Assignment
  - How to get started
  - How to submit your FA
- What does a good business proposal look like?
  - How does it apply to our FA?
- FAQ

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(Recap) Course Completion Requirements

You will be evaluated on the following assignments:

	Total	Completed	Honors	Outstanding
Attendance Surveys	14	7+	7+	Selected Students (based on overall scores)
Homework	24 points (=8 HW) *3 points max, 2 points if late	14+ points	14+ points	
Final Assignment	1	Pass	TOP 10%	
Competition	1	(optional)	TOP 20%	

You need to submit & pass Final Assignment to complete the course.

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Final Assignment

- You are required to make presentation slides to propose a business strategy using the provided datasets.

- Deadline: **November 28th, 11:00 am UTC**

- Please submit
  - your presentation slides,
  - Jupyter notebook file(s) that you used for your analysis, and
  - your list of references.

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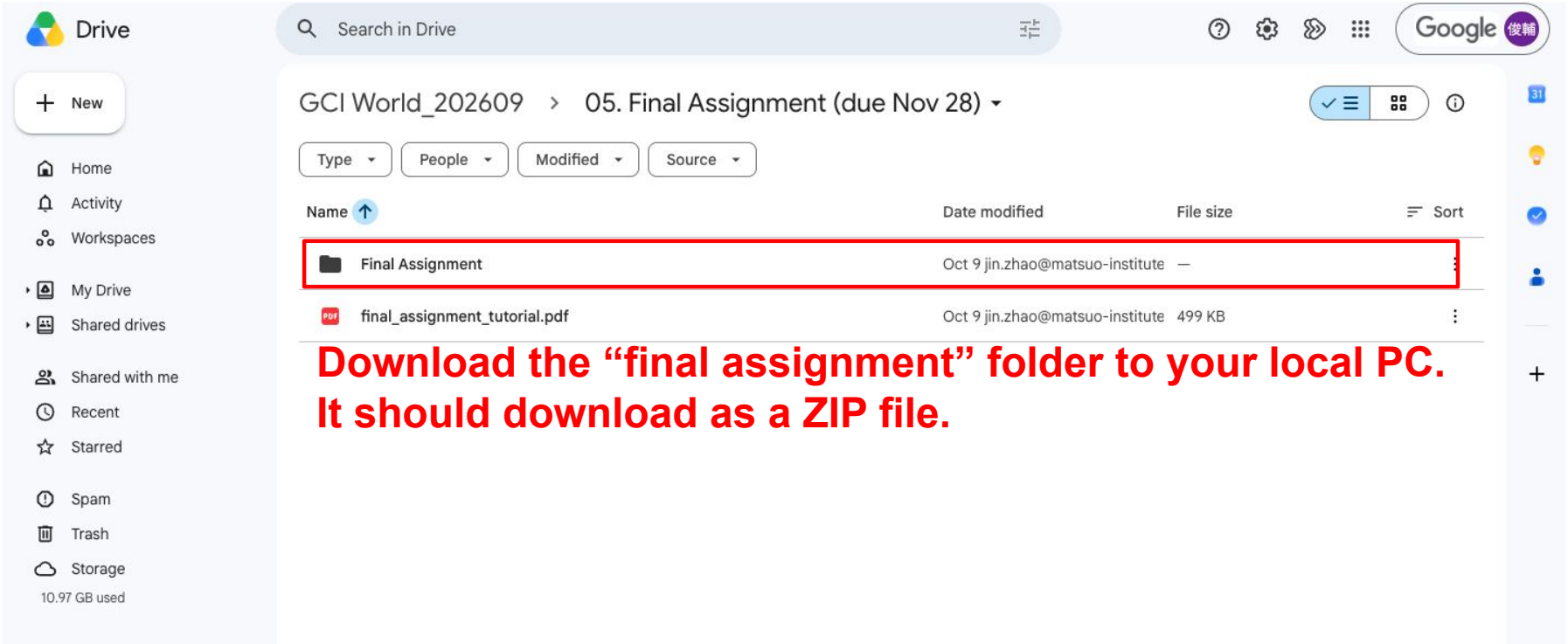
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1. Download the materials

Download the “Final Assignment” folder from the class materials



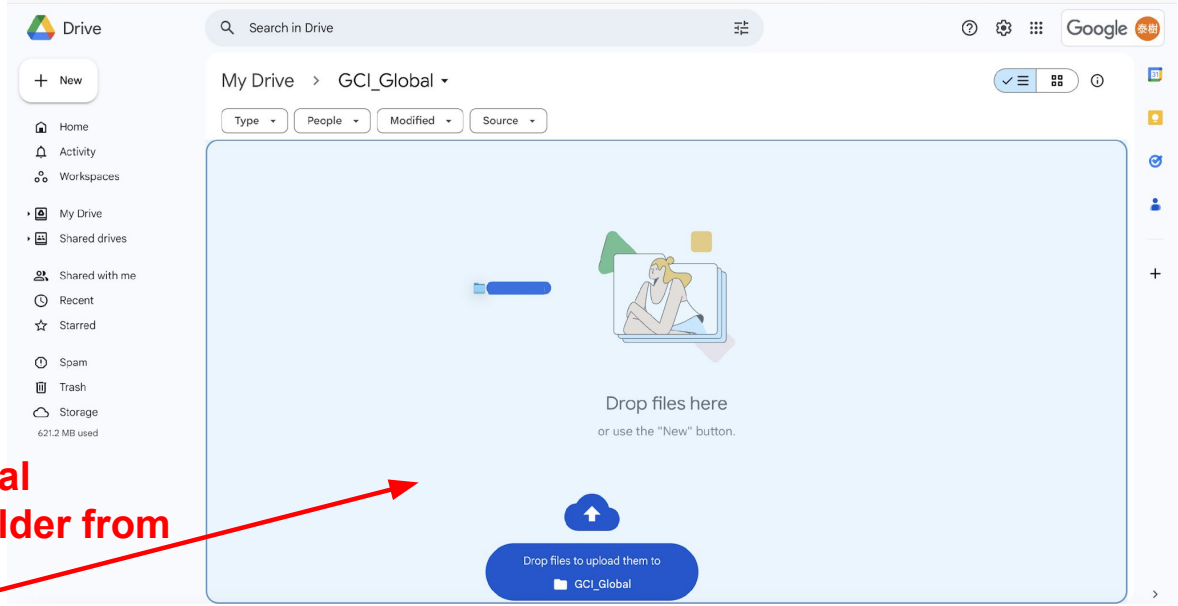
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2. Upload the dataset to your Google Drive

Unzip the downloaded ZIP file.

Then, upload the “Final Assignment” folder to your MyDrive.

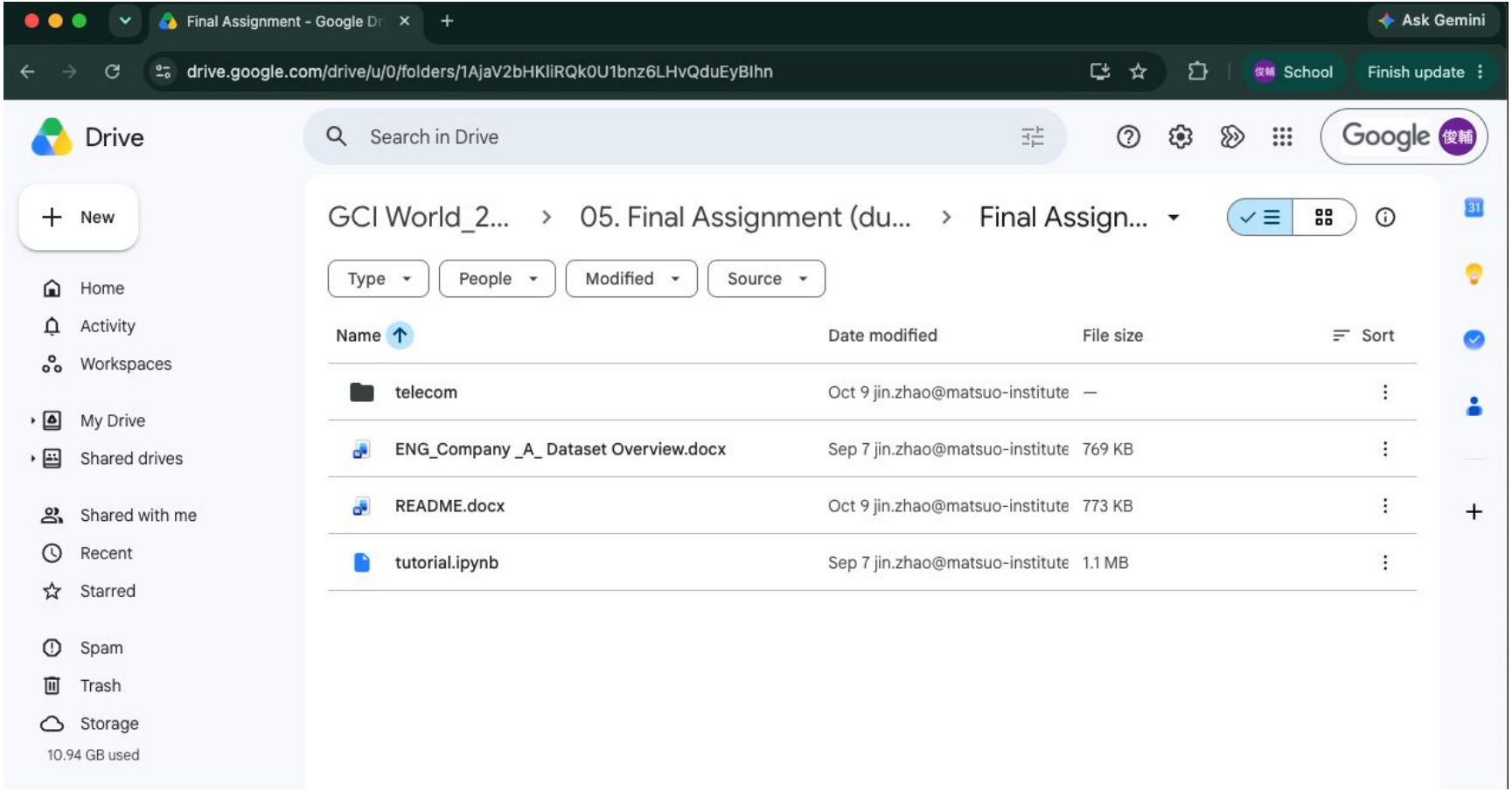
(It can be anywhere in your MyDrive, but you will need to adjust the PATH accordingly later)



Upload the “Final Assignment” folder from your local PC

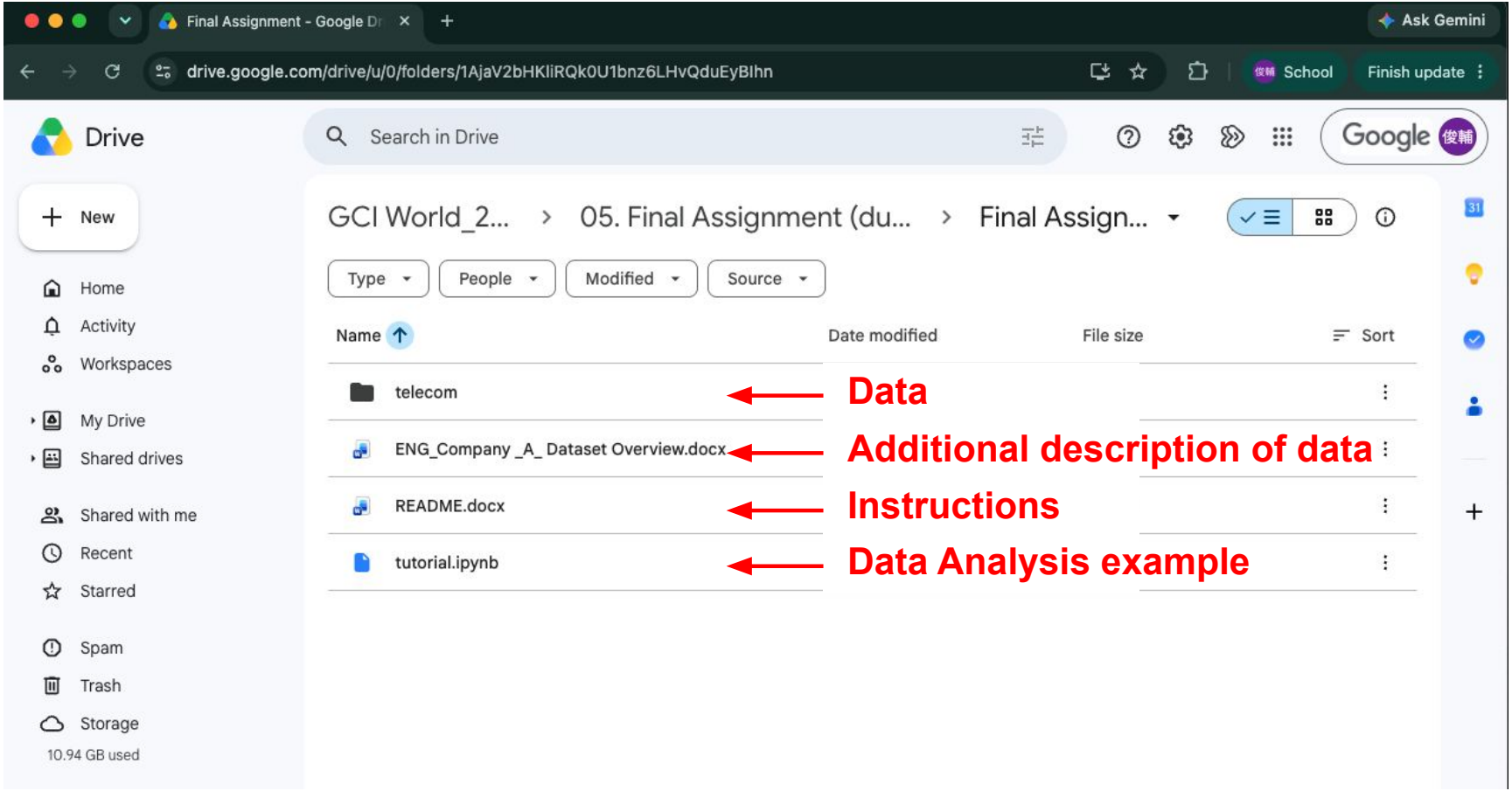
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Final Assignment folder



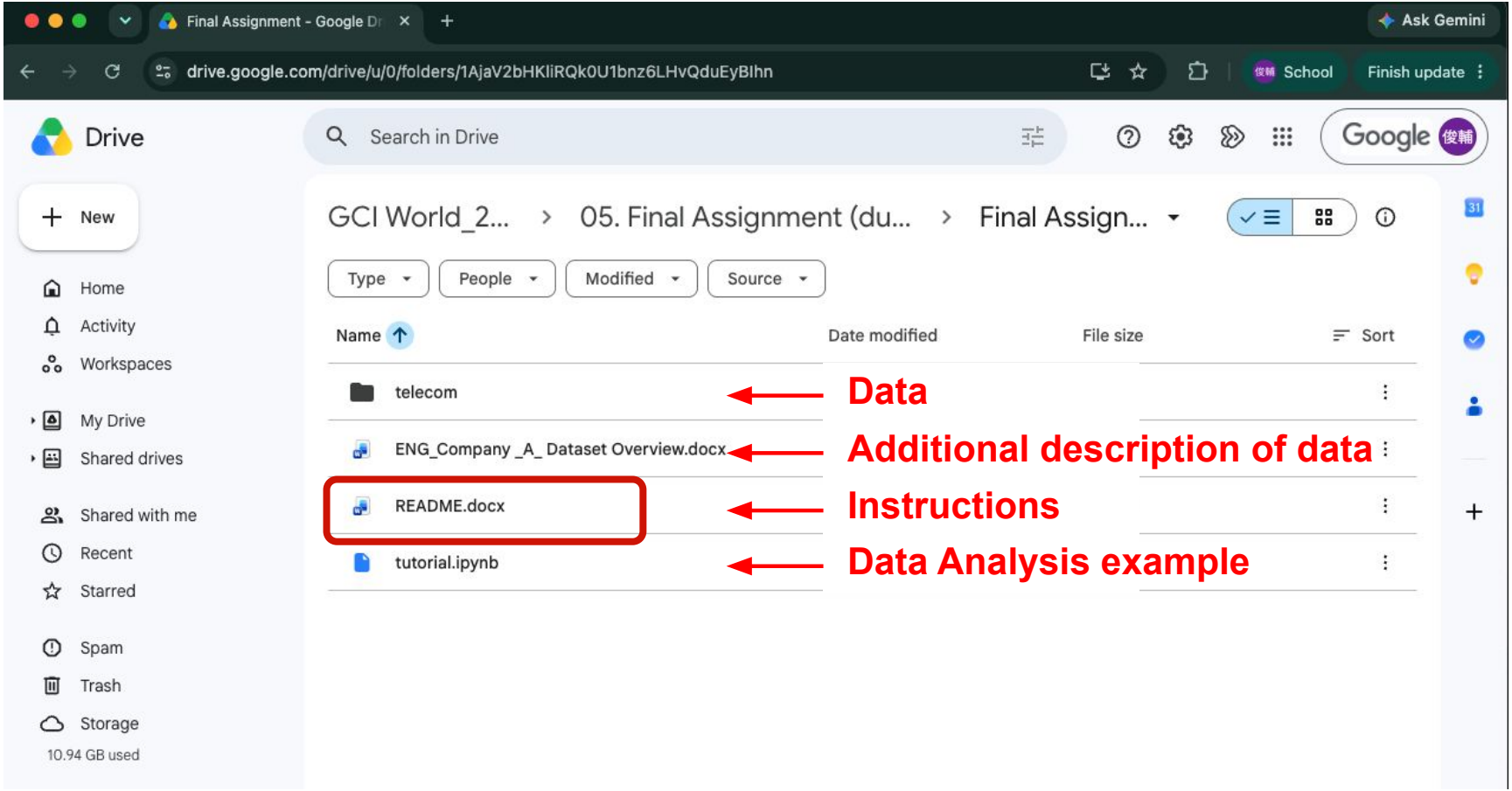
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Final Assignment folder



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Final Assignment folder



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README

1. Scenario & Objective

You are a new associate at an IT consulting firm. The sales team has just secured a PoC (Proof of Concept)* project from a company. The company has a large dataset but has limited in-house expertise to utilize them for business. **Your job is to run EDA and build at least one machine learning model on the dataset. From evidence and market analysis, you need to define the client's core business problem. Then, you will create a data-driven business proposal with quantified impact (e.g. revenue lift, cost savings, risk reduction, etc.)**.
As an associate, your proposal must be data-driven and machine learning-based.
A strong proposal may lead to a formal consulting contract and give you the opportunity to lead a machine learning–based business development project.

***Proof of Concept (PoC)**: an experimental verification process conducted before fully implementing a new technology or idea. Its purpose is to confirm feasibility and requirements. If successful, the project may advance to full-scale system development or business strategy formulation.

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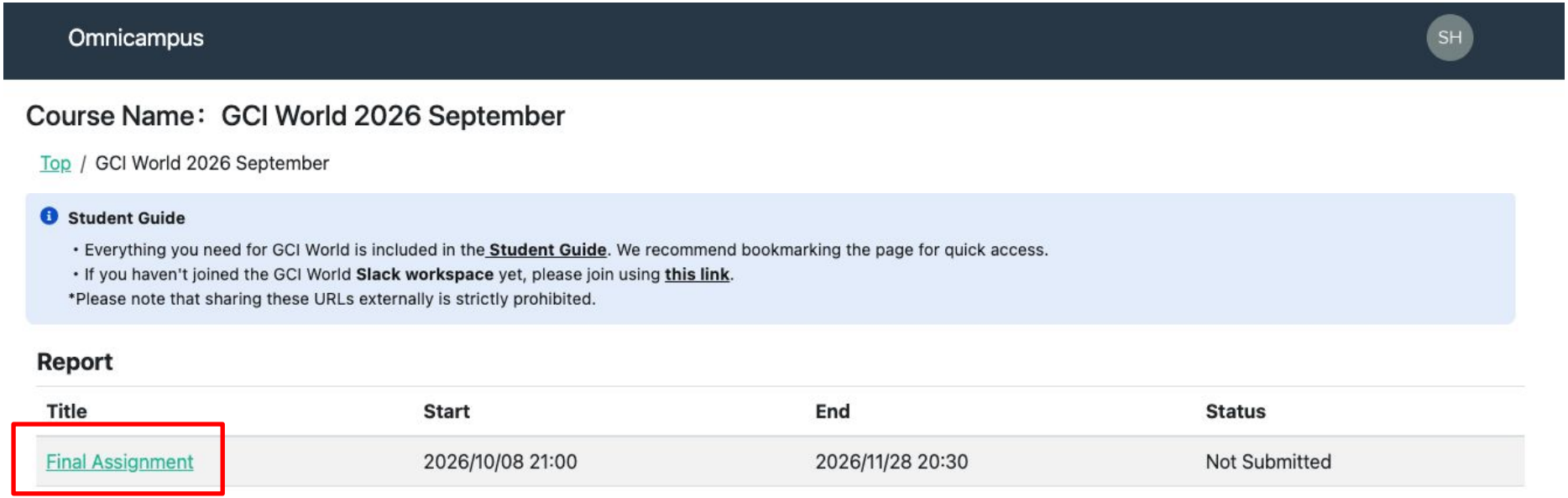
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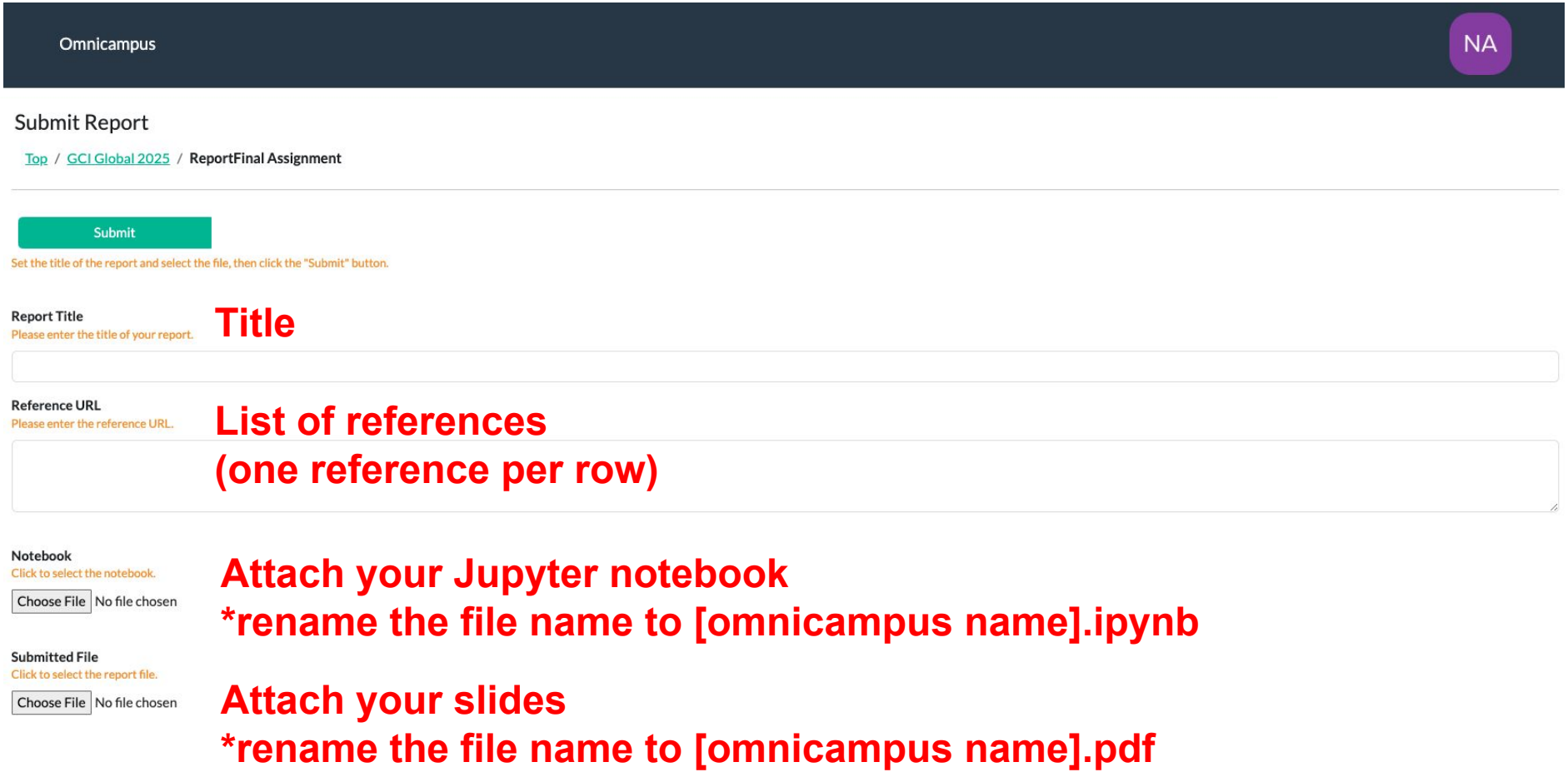
5-1. Select the assignment

Login to Omnicampus and select the assignment.



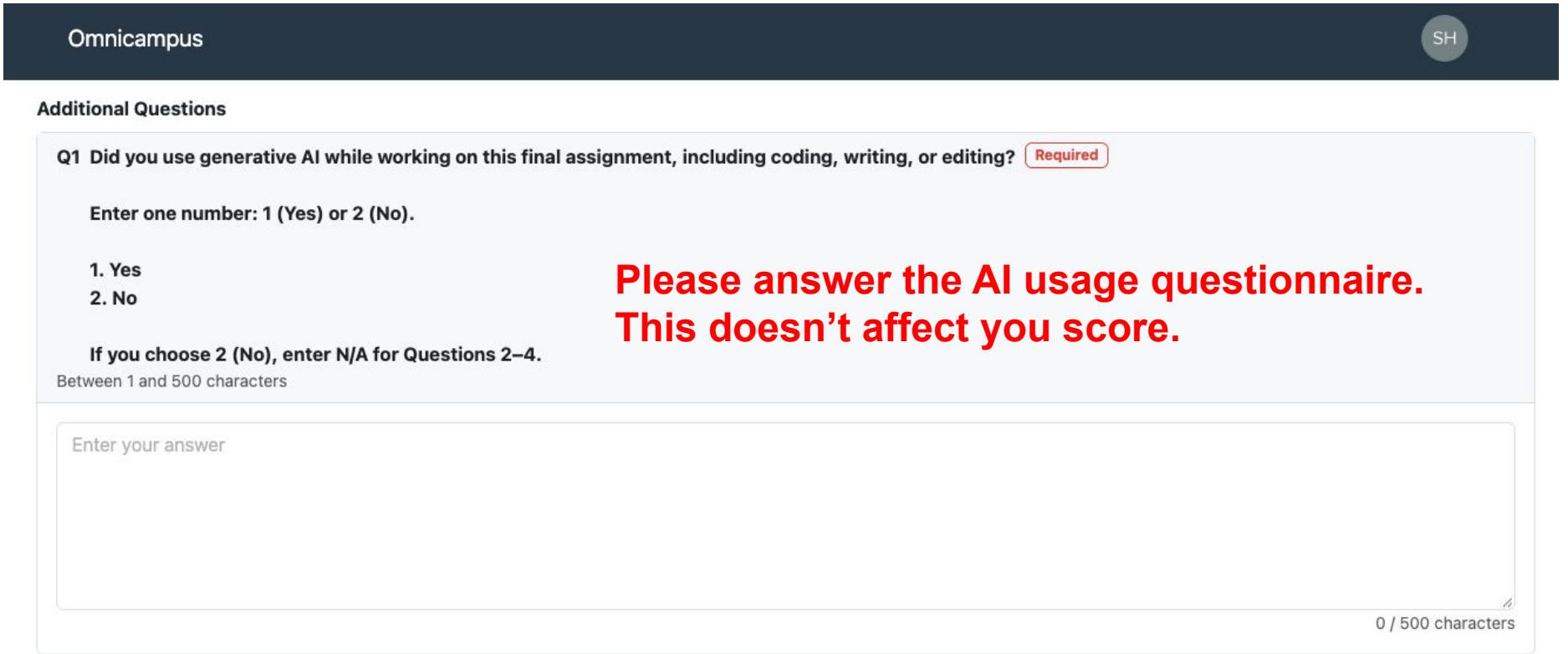
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5-2. Attach the files (make sure to rename the files)



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5-3. Fill in the AI usage questionnaire



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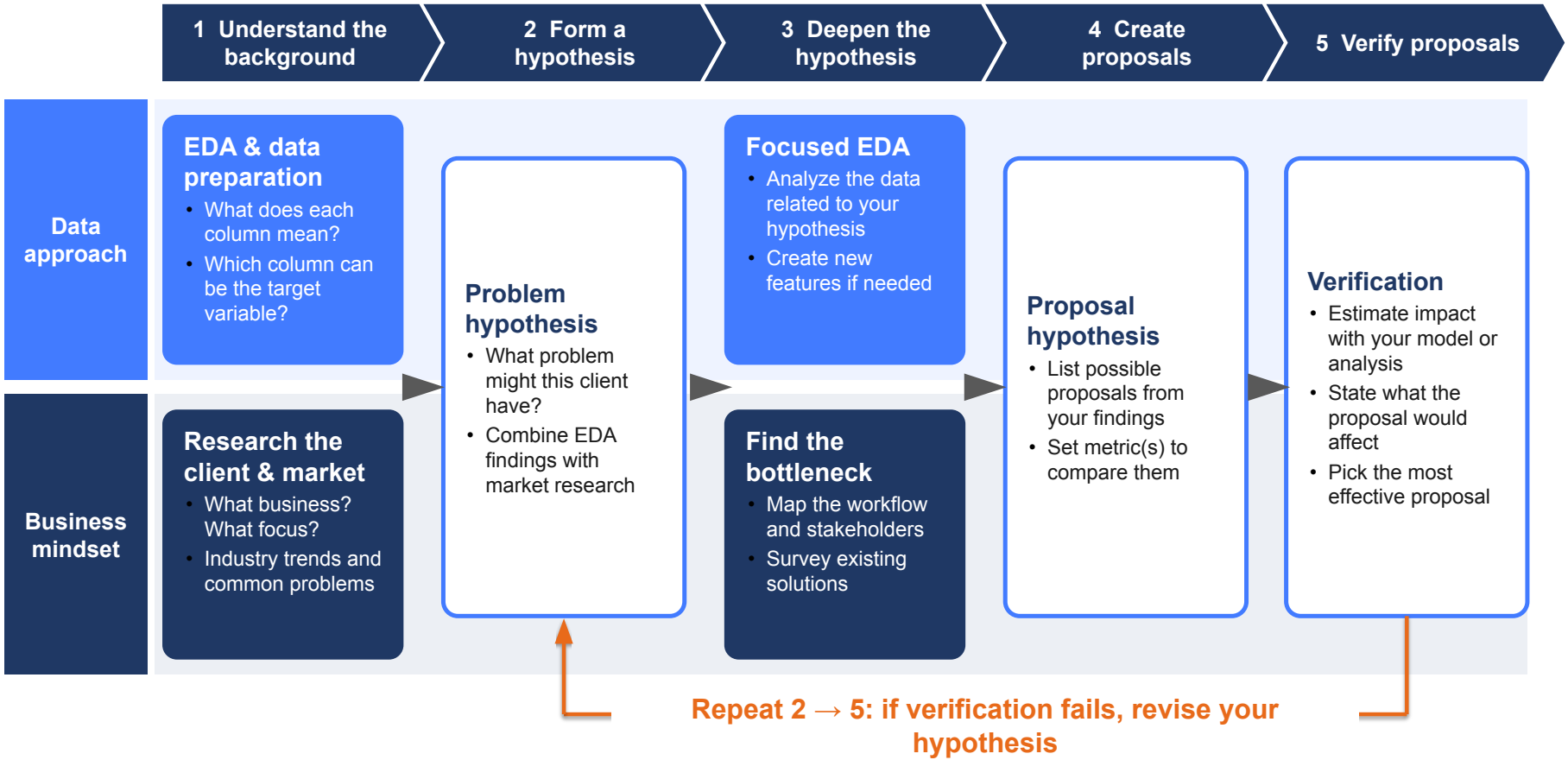
Your task is to

- create a **business proposal based on the given data**.
- **identify the client's business problem and present a solution**.

What does a good business proposal look like, in the first place?

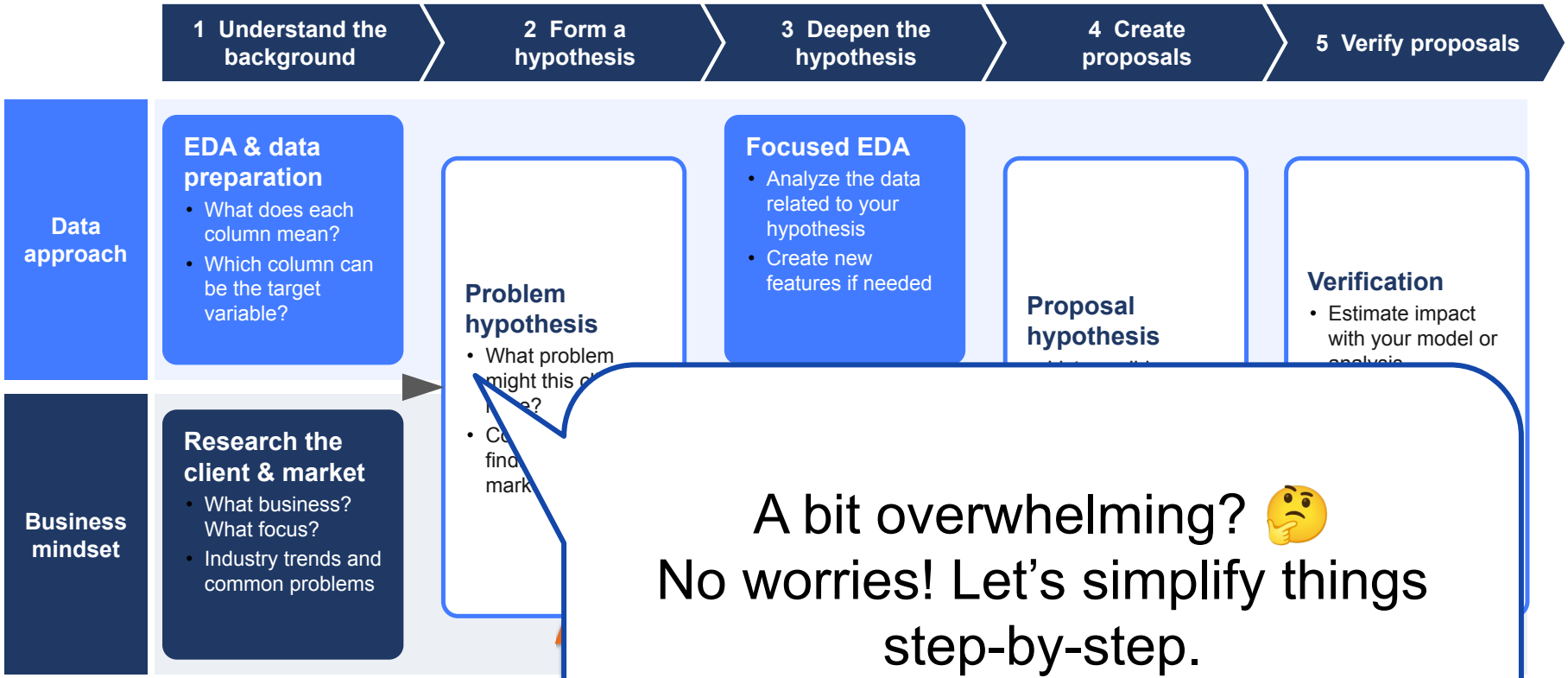
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Business proposal: Workflow



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Business proposal: Workflow



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Three key points: good business proposal

A good business proposal

1. specifies an important issue worth solving.

2. includes a hypothesis to solve that issue.

3. is verifiable.

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Three key points: good business proposal

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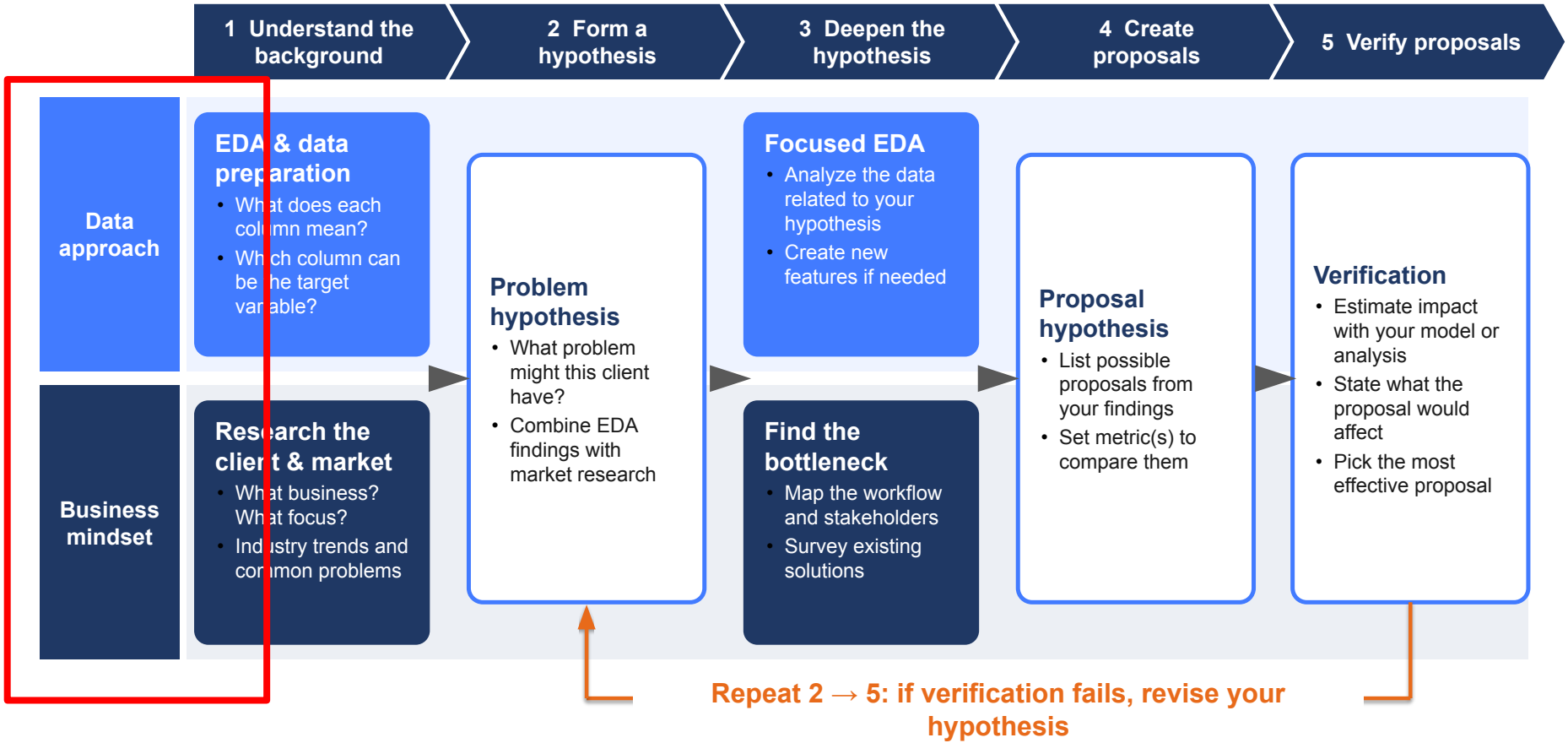
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Business proposal: Workflow



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1. Settle your issue



Client

Our product A is suffering from sluggish sales, although it has more functions, more settings, and better performance than its competitors. What are we going to do about this?

The performance of the product is already great. The marketing strategy seems to have a problem. Let's start from running more web advertising!

You

Are you referring to the real issue?

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1. Settle your issue



Client

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1. Settle your issue

"Sluggish sales":

- How big is the market?
- Do other products also face sluggish sales?

Market background analysis

"More functions":

- Are customers happy with more functions?
- Doesn't it worsen its operability?

Customer survey

"Better performance":

- Is it necessary to have better performance?
- Isn't it just enough to have less?

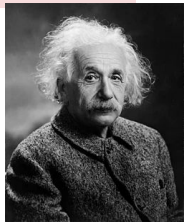
Another product may cover up the sales

You must find and focus on the "core" issue.

Collecting **background information** and some **quantitative analysis** (**EDA**) help you find an issue worth working on!

"The formulation of a problem is often more essential than its solution, which may be merely a matter of mathematical or experimental skills."

Albert Einstein



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Three key points: good business proposal

A good business proposal

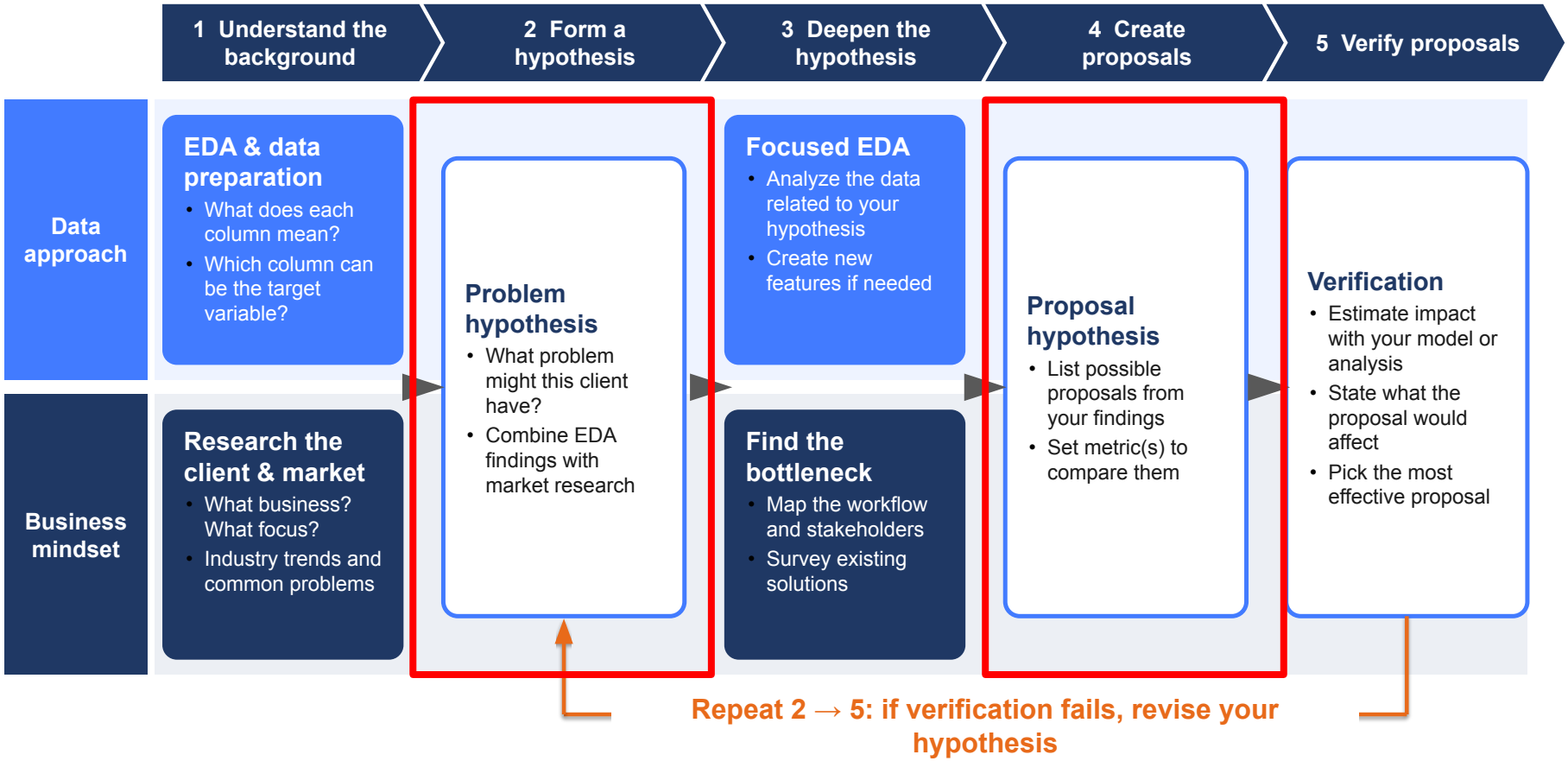
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2. Build your hypothesis

Once you find an issue, you should settle your **hypothesis** on the issue.

What is a hypothesis?

A **hypothesis** is a tentative answer to your issue that seems the closest to the correct answer.

Why is it critical?

- Without focusing on a hypothesis, you cannot take action or make clear decisions.
  - The lack of a hypothesis **paralyzes people** especially when they are given only data.
- The initial hypothesis can be wrong – the process needs repetitive elaboration.

How to make a good hypothesis?

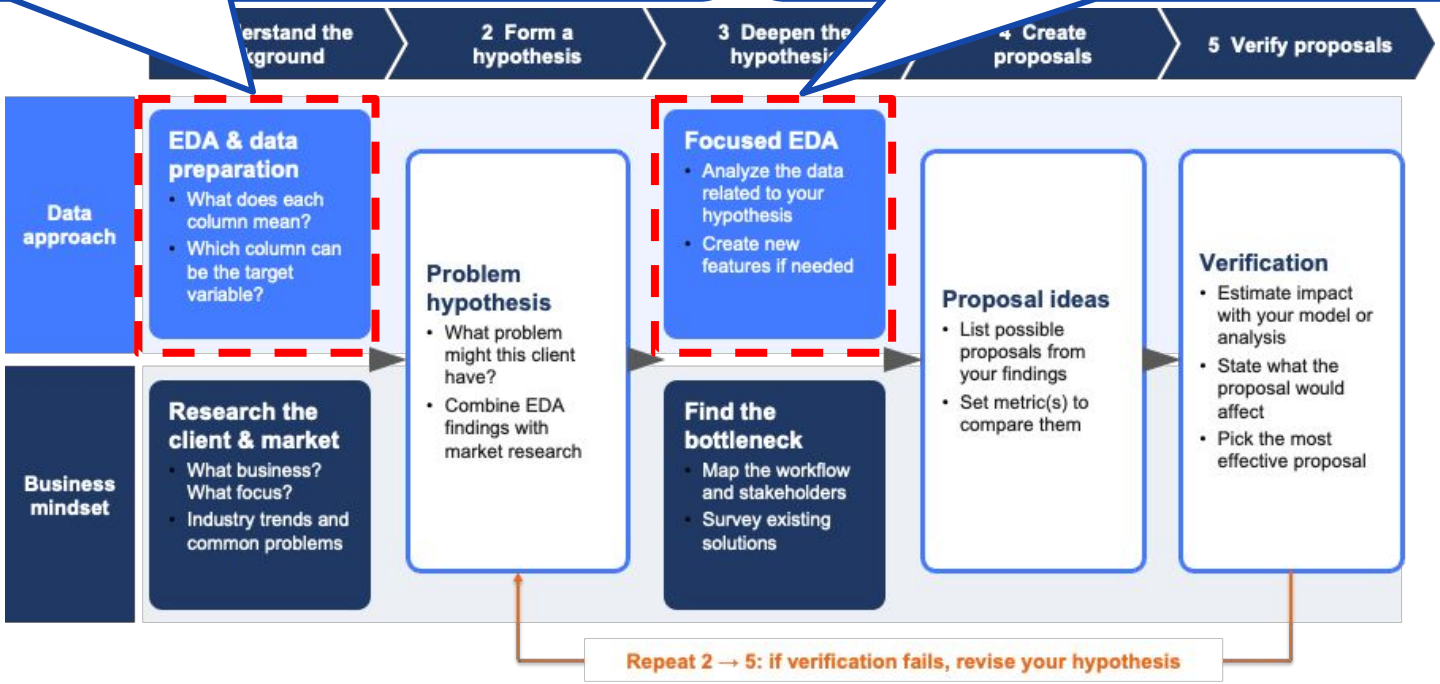
- This is where your **data analysis skills** come in!
  - Good hypothesis stands on quantitative analysis and insights based on it.
- In the real business settings, you can also collect real opinion from the people involved.
  - Customers, salespeople, manufacturers, personnels in other departments, etc.

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Two types of EDAs

① EDA & data analysis to determine your issue & initial hypothesis
- Can be **explorative**, but **not too haphazard**
- Grasp an overview and insight

② EDA to deepen your hypothesis
- More **focused**
- Try to answer your initial hypothesis
- Use concrete methods, **ML models**, etc.



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Three key points: good business proposal

A good business proposal

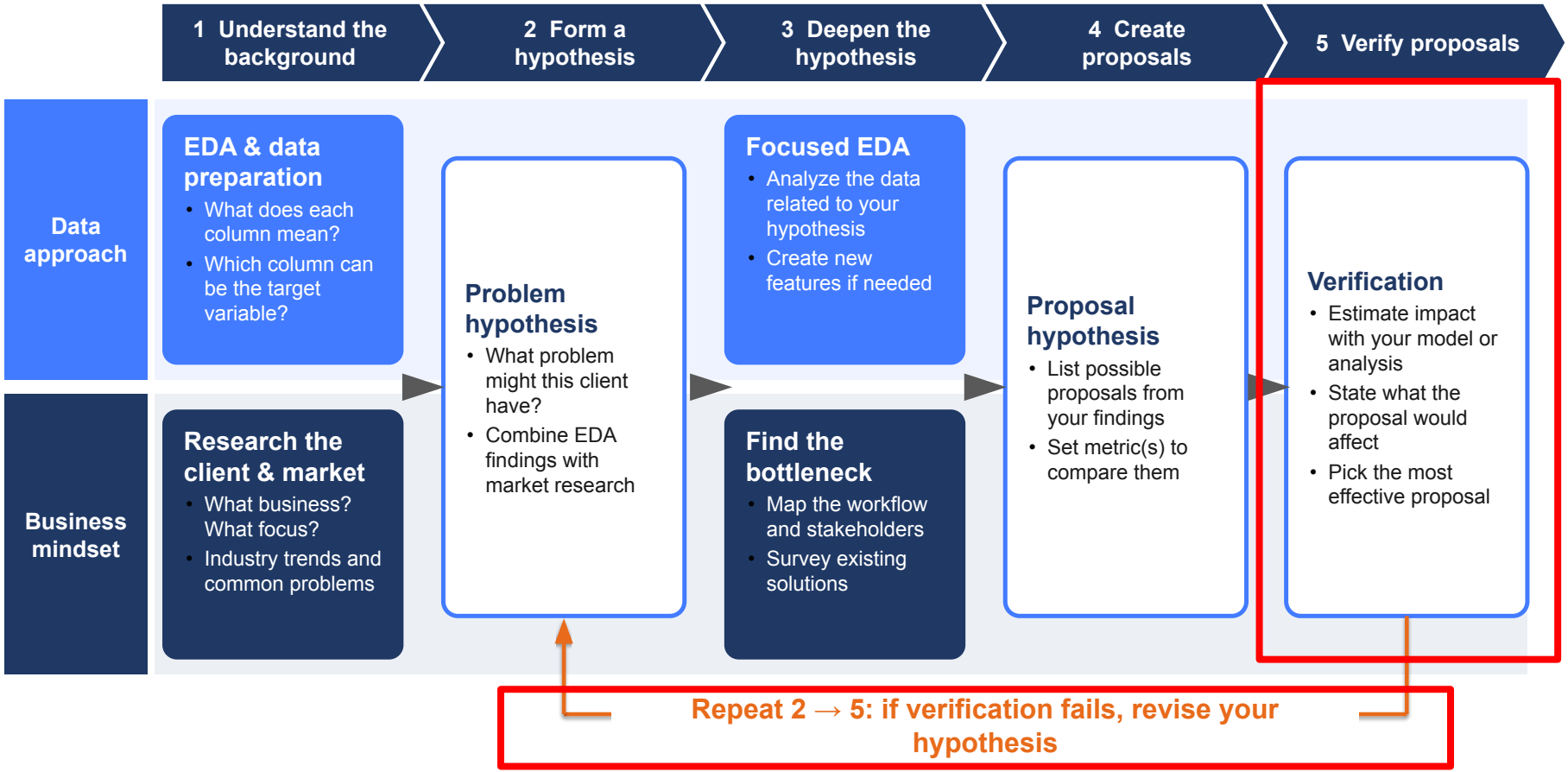
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3. Last but not least, make it verifiable

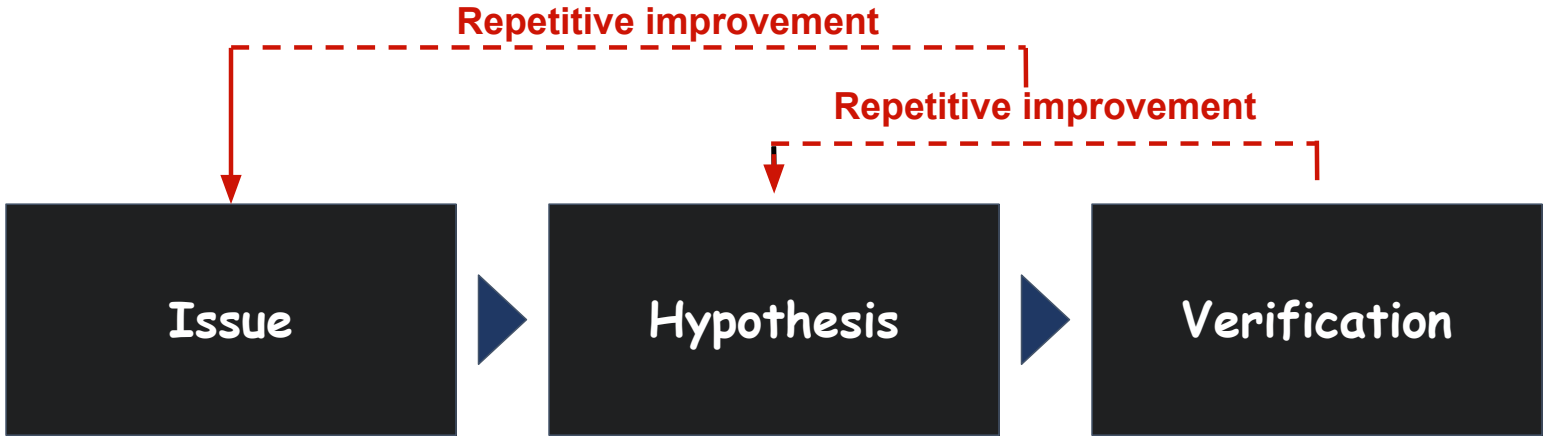
Specifying your core issue and building your hypothesis and proposal is not the end: you need to verify & validate of your proposal.

You need to answer questions such as:

- What the **expected profit** of your proposal?
- What is the **limitation**? No proposal on the earth is perfect.
- How do you **continuously test & verify** your proposal? How do you reflect feedbacks from customers?

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Cycle of making your Issue, hypothesis, and verification



The question to be addressed
- influences the whole later directions

Tentative answer
- concrete, verifiable, actionable

Appropriate steps according to the hypothesis
- Data analysis
- Literature review
- Experiments
- Consumer survey etc.

This repetitive cycle leads us to good issues hypotheses.
Gathering that information is also important since arguments and hypotheses change depending on the background information.

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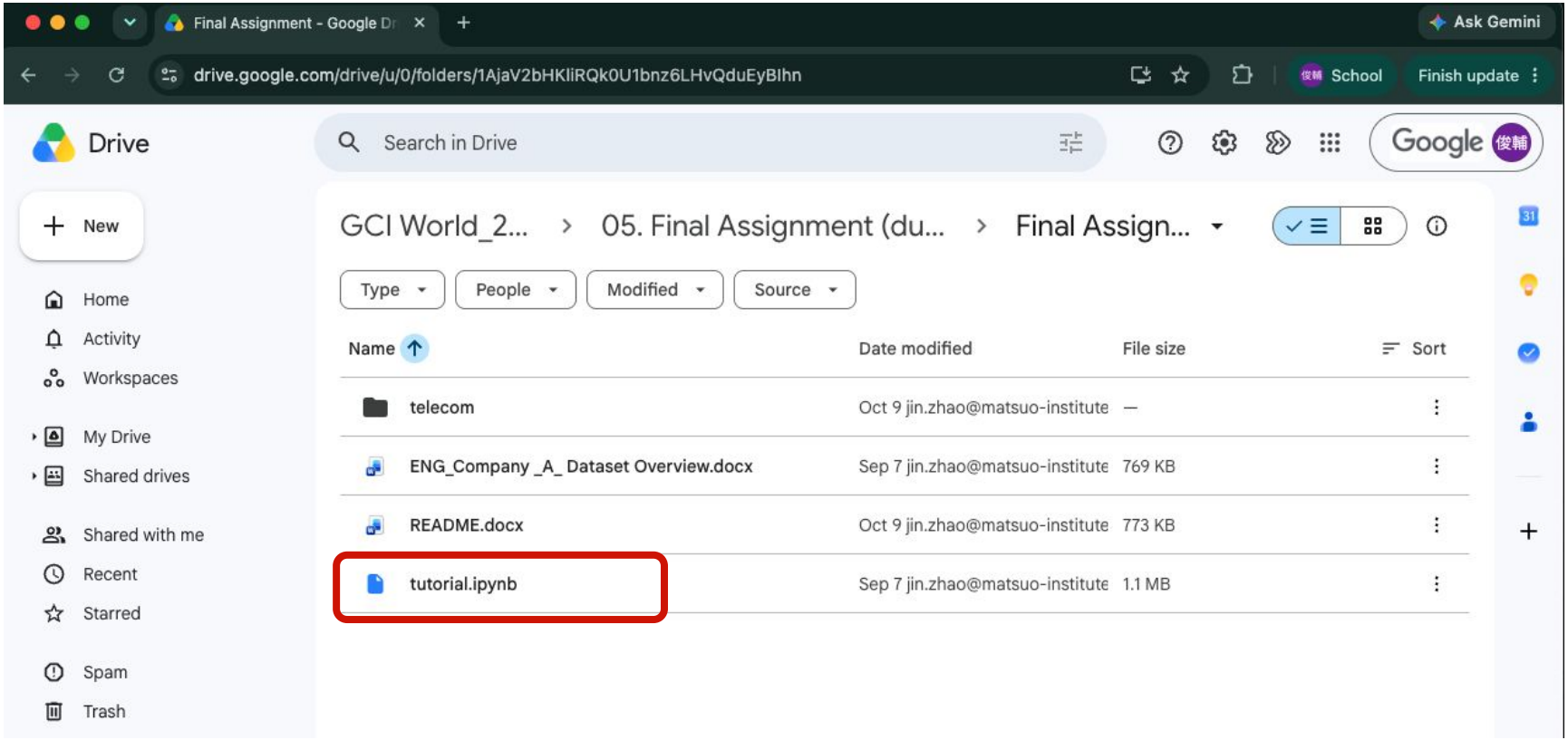
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How today's instructions apply to your Final Assignment

- Please go to the `tutorial.ipynb` notebook



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How today's instructions apply to your Final Assignment

- The most straightforward **issue** may be "**how to increase the profit by decreasing the churn rate?**"
  - There can be, however, other choices of issues to approach.
  - Your first EDA and background knowledge survey would give you hints of what issues are there.
- After some EDA, you can gain some insights on what your **hypothesis** should look like.
  - Do longer equipment ages increase the churn ratio?
  - Can you explain why your hypothesis looks persuasive by additional more focused EDAs?

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How today's instructions apply to your Final Assignment

- How do you make your proposal verifiable?
  - In Final Assignment, you can't receive the real feedback of your proposal. But at least you have to show
    - what makes your proposal legitimate.
    - the **projected profit** of your proposal.
    - the way to **continuously test, verify, and update** your proposal.
  - What is the limitation? No proposal on the earth is perfect. Can you deal with the limitation in your verification cycle?

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Example of the past Outstanding Students

Will be published in “05. Final Assignment” folder soon!

Introduction of Data

Approximately 200 types of customer information have been accumulated as data.

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	...
0	100002	1	Cash loans	M	N	Y	0	202500.0	406597.5	24700.5	...

	SK_ID_PREV	SK_ID_CURR	NAME_CONTRACT_TYPE	AMT_ANNUITY	AMT_APPLICATION	AMT_CREDIT	AMT_DOWN_PAYMENT	AMT_GOODS_PRICE	WEEKDAY_APPR_PROCESS_START
0	2030495	271877	Consumer loans	1730.430	17145.0	17145.0	0.0	17145.0	SATURDAY
1	2802425	108129							
2	2523466	122040							
3	2819243	176158							
4	1784265	202054							

	SK_ID_CURR	SK_ID_BUREAU	CREDIT_ACTIVE	CREDIT_CURRENCY	DAYS_CREDIT	CREDIT_DAY_OVERDUE	DAYS_CREDIT_ENDDATE	DAYS_ENDDATE_FACT	AMT_CREDIT_MAX_OVERDUE
0	215354	5714462	Closed	currency 1	-497	0	-153.0	-153.0	NaN
1	215354	5714463	Active	currency 1	-208	0	1075.0	NaN	NaN
2	215354	5714464	Active	currency 1	-203	0	528.0	NaN	NaN
3	215354	5714465	Active	currency 1	-203	0	NaN	NaN	NaN
4	215354	5714466	Active	currency 1	-629	0	1197.0	NaN	77674.5

A portion of the data you provided

Data Contents

- Key data for each customer (gender, occupation, family information, presence/absence of default, etc.)
- Information on customers' past borrowings at other financial institutions (delinquent amounts, credit limits, etc.)
- Monthly balance status of customers
- Credit card usage information and withdrawal amounts for the month prior to the application date
- Key data at the time of past applications (contract type, application amount, purpose of use)
- POS cash balance information
- Installment payment information

Proportion of Non-Defaulters



n= 307506

4

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FAQ (1/4)

Q Am I going to fail Final Assignment even if I submit my report, if the quality is too bad?

A Yes. We have our own criteria for grading Final Assignment.
Just submitting your FA is not enough for completion. You must meet our requirements to pass FA, eventually to complete our course.

Q There should be so many submissions. Do you use AI for grading?

A We are afraid we can’t answer anything about the concrete process of grading.

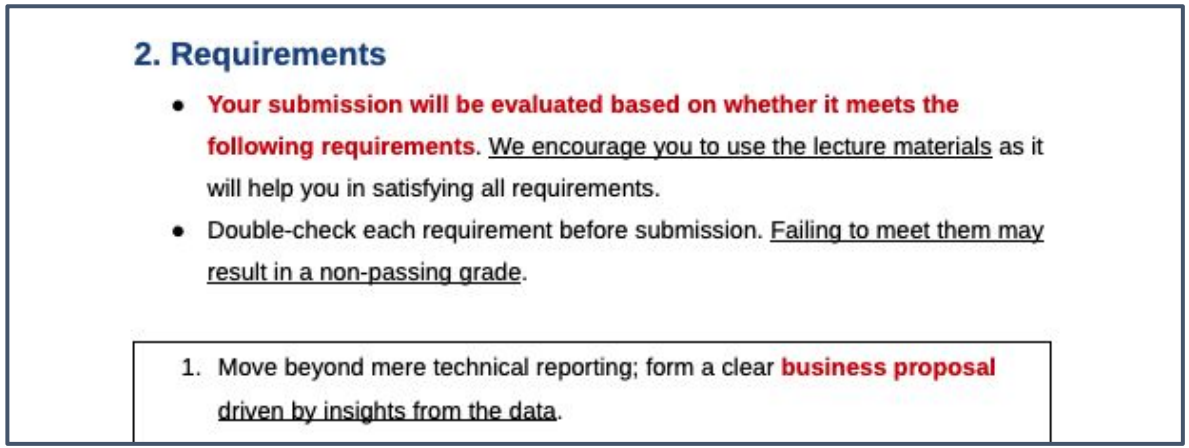
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FAQ (2/4)

Q We want to know the grading rubrics.

A We are afraid that we are not planning to release our rubrics.

But in fact, the **“2. Requirements” section in README tells you pretty much** about how your report is going to be graded. So we strongly recommend you to **read it carefully & thoroughly and follow the requirements**.



2. Requirements

- **Your submission will be evaluated based on whether it meets the following requirements.** We encourage you to use the lecture materials as it will help you in satisfying all requirements.
- Double-check each requirement before submission. Failing to meet them may result in a non-passing grade.

1. Move beyond mere technical reporting; form a clear **business proposal** driven by insights from the data.

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FAQ (3/4)

Q Can I use AI for my FA?

A **Yes**. **We actually recommend the use of AI, which accelerates your hypothesis-evaluation cycle**. Any AI usage will not affect your score. But if you use any types of generative AIs, please share the URL to the output/chat history.

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Additional notes about AI Usage

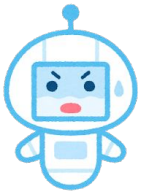
A **Using AI doesn't necessarily decrease your workload**!

AI helps you dramatically in

- Providing methodologies, implementing code, and visualizing.

But, **AIs still have problems in**

- **specifying core issues** with the big picture,
- **building nice hypothesis**,
- or **making presentations easy for humans to understand**.



AIs in no doubt accelerate your hypothesis verification cycle.
AIs, however, **tend to lack comprehensive and broader perspective** on a problem, often being **too particular about insignificant details.**

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FAQ (4/4)

Q Can you extend the deadline? I had my school exam.

A **No**. We are afraid but **we will never make a deadline extension**. Please make sure you can submit before the deadline. **No school schedule, technical difficulties, or other personal situations can be your excuse**.

Please submit your report with plenty of time to spare!

Q Can I make my FA in my own language?

A **No**. Please use **English** in your presentation.

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Other Important Notes

- Final Assignment should be in **presentation slides ≦ 15 pages**, **not in document**!
- Make it readable to those who first read your report.
  - Make it clear how each slide is connected to the previous/next one.
  - No obscure wordings, no unexplained jargons (LLMs always make them!).
  - No too much decorations: reader-friendliness and clarity matter much more.

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Other Important Notes

- Put figures in your presentation slides, not just in your notebook.
- Don’t change your theme – i.e., don’t try to solve your own problem.
  - The presentation should be based on what we provided.
- **Again, we will never make a deadline extension**, so please make sure you can submit before the deadline. Submit with plenty of time to spare!

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References of this section

- Kei-ichi Yokoyama, Presentation for Session 10 in GCI Winter 2024 (Japanese domestic course)
- Kazuto Ataka, “Issue Driven” (Japanese)
- Kazunari Uchida, “The BCG Way – the Art of Focusing on the Central Issue” (Japanese)
- Thomas Wedell-Wedellsborg, “What’s Your Problem?”

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Q&A session for Final Assignment

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